

Cainiao is Fast — But Not Everywhere

When does Alibaba’s logistics network stop delivering its speed advantage?

BUAD 345: Decision Analytics and Visualization
Final Project

Summer 2026

Audience: Cainiao (Alibaba’s logistics platform). **Data:** Tmall order, logistics, and merchant records, Jan 1–Jul 31, 2017 (1,291,197 orders; 1,288,438 with a complete, valid fulfillment time).

1. Motivation

In Chinese e-commerce, delivery speed is a primary driver of customer satisfaction and repeat purchase, and it is the metric on which logistics platforms compete most visibly.¹ Cainiao’s value proposition is that its smart-routing platform makes fulfillment *faster*. In our data Cainiao handles **30.2%** of all orders, so any systematic weakness in its speed advantage is material to the platform.

2. Main Idea

We test one focused question: *does Cainiao actually make order fulfillment faster, and if so, is that advantage uniform?* Our outcome is total fulfillment time, defined exactly as in the course materials:

$$\text{totaltime} = \text{SIGNED timestamp} - \text{pay_timestamp} \quad (\text{hours}).$$

We then dig *hierarchically*: carrier size, then destination-city size, then a segment-level decomposition of where the time is spent.

3. Findings

3.1 Overall, Cainiao is faster. Cainiao orders are fulfilled in **61.9 h** on average versus **71.0 h** for non-Cainiao orders — about **9.2 hours faster** (medians: 53.7 vs 66.3 h). The gap survives a fair-comparison control: restricting to orders with *no* promised-speed window (`promise = 0`) leaves it essentially unchanged (62.5 vs 71.0 h). See Figure 1.

3.2 The advantage is uneven across carriers. Splitting by carrier size (“Large” = a carrier handling >200,000 orders; the two largest carriers cover the bulk of volume), Cainiao’s edge is clearly larger on large carriers than on small ones.

3.3 Punchline: big-city destinations on small carriers. Adding destination-city size (Big = the two largest destination cities) exposes the real story. Cainiao saves 8–10.5 hours in three of the four cells — but in **big-city destinations handled by small carriers, Cainiao is actually 3.3 hours slower** than non-Cainiao (Table 1, Figure 2).

¹Replace with an assigned reference, e.g. a Cainiao/Alizila press item or an operations-management study on delivery speed and customer retention.

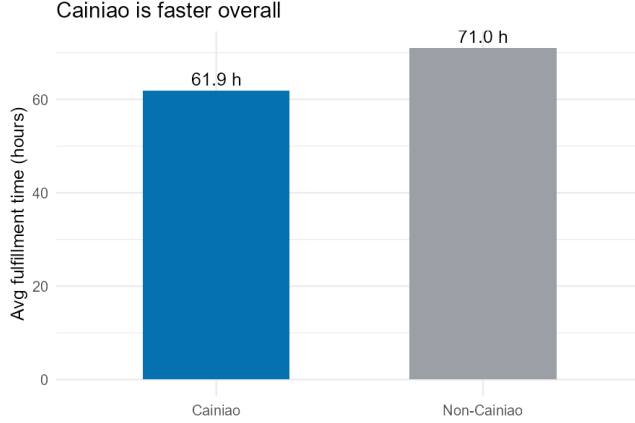


Figure 1: Cainiao is faster overall.

Table 1: Average fulfillment time (hours) and Cainiao’s advantage by cell.

Destination	Carrier	Cainiao	Non-Cainiao	Cainiao hours saved
Big city	Large	44.7	54.2	+9.5
Big city	Small	52.4	49.1	-3.3
Small city	Large	61.7	72.2	+10.5
Small city	Small	67.1	75.2	+8.0

Customer reviews echo it. The same cell is Cainiao’s weakest on logistics review score (4.764, its lowest of any cell, versus non-Cainiao’s 4.860, the highest), so customers feel the slowdown too (Figure 4).

4. Mechanism: where does the time go?

Decomposing fulfillment time into pre-delivery (order → pickup), line-haul (between-facility transit), and last-mile (final station → buyer) shows the gap lives in **line-haul, not the last mile**. Last-mile is small and similar across all groups (roughly 4–6 h). Cainiao’s usual edge comes from much faster line-haul (e.g. 38.4 h vs 47.8 h on small-city/large-carrier orders) — but in the punchline cell that line-haul advantage disappears (27.0 h vs 26.9 h, essentially tied). See Figure 3.

An honest non-result. A natural hypothesis — that small carriers make more consolidation stops — is *not* supported: small carriers pass through slightly *fewer* cities on average (about 2.9–3.1) than large ones (3.4–3.7). So the penalty is per-leg transit/dwell time, not a larger number of hops.

5. Potential Reasons

Two domain explanations are consistent with the pattern. First, big cities concentrate volume and inter-city trunk lines; large carriers run dedicated, high-frequency line-haul into them, while small carriers must wait to consolidate loads, adding transit/dwell time precisely on the high-expectation big-city lanes.² Second, small carriers likely lack the capacity to execute Cainiao’s smart-routing

²Replace with an assigned reference on Chinese logistics infrastructure / carrier line-haul economics.

...BUT the advantage shrinks for Big-city destinations on Small carriers

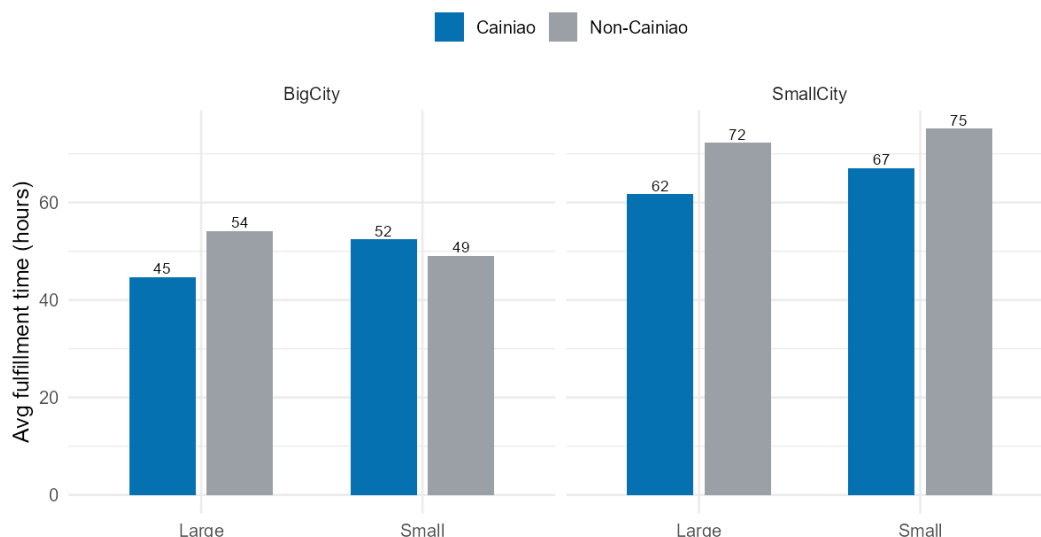


Figure 2: Cainiao’s speed advantage nearly vanishes (and reverses) for big-city destinations served by small carriers.

recommendations as designed, so the platform’s optimization does not translate into time saved for them.³

6. Practical Implications (So What?)

For Cainiao:

1. **Deepen ties with large carriers** on big-city lanes for reliable speed — while monitoring the concentration risk this creates.
2. **Make smart-routing carrier-capability-aware** (size, coverage, experience) so small carriers are not assigned routes they cannot execute efficiently.
3. **Pilot big-city consolidation / pickup lockers for small-carrier shipments** to attack the line-haul and pickup penalty where it actually occurs.⁴

7. Limitations & Causality

This is observational data: assignment to Cainiao is not random, so differences reflect association, not proven causation (e.g. merchants choosing Cainiao may differ systematically). Milestone records are missing for some orders in this real-world feed, so segment means use slightly different subsets than the totals. The big-city flag is a coarse two-city proxy. These caveats temper the magnitudes but not the qualitative punchline, which is large and consistent across time, review scores, and segments.

³Replace with an assigned reference on routing-algorithm performance under heterogeneous carrier capability.

⁴Replace with an assigned reference on pickup-point / locker effects on last-mile and consolidation efficiency.

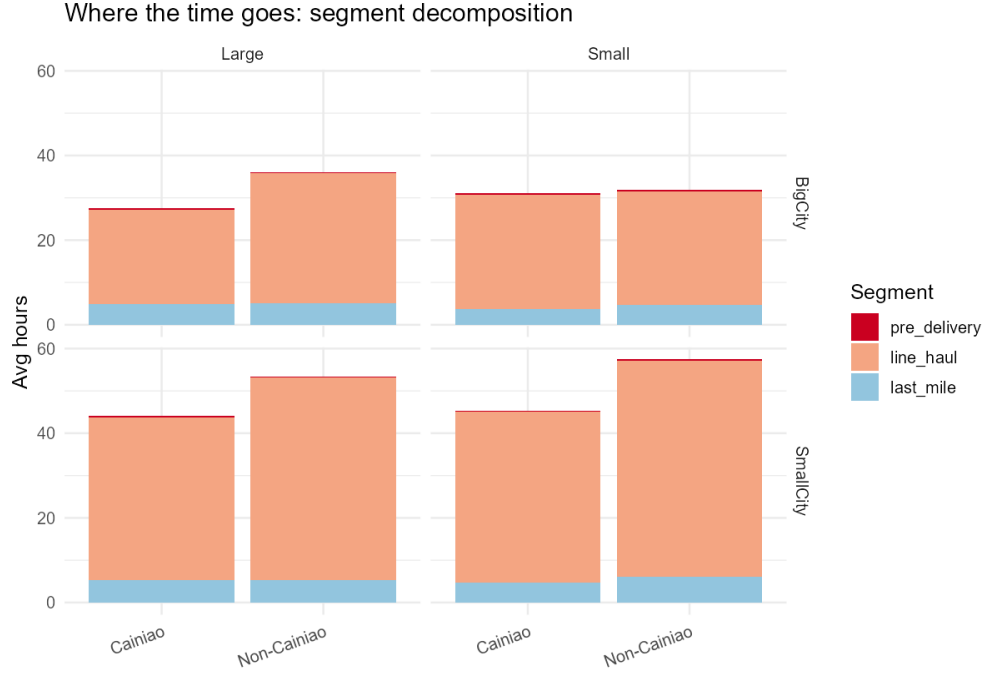


Figure 3: Segment decomposition. The between-facility line-haul leg dominates and is where Cainiao normally wins; that win evaporates for big-city/small-carrier orders. (Segment means are computed over orders with complete milestones and are not strictly additive to the totals in Table 1.)

Two-Sentence Summary

(1) Cainiao significantly shortens order fulfillment time across the marketplace (about 9 hours on average). (2) But that advantage largely disappears — and even reverses — for big-city destinations served by small carriers, with the bottleneck in between-facility line-haul rather than the last mile.

Customer reviews echo it: Cainiao's lowest score is Big-city x Small carrier

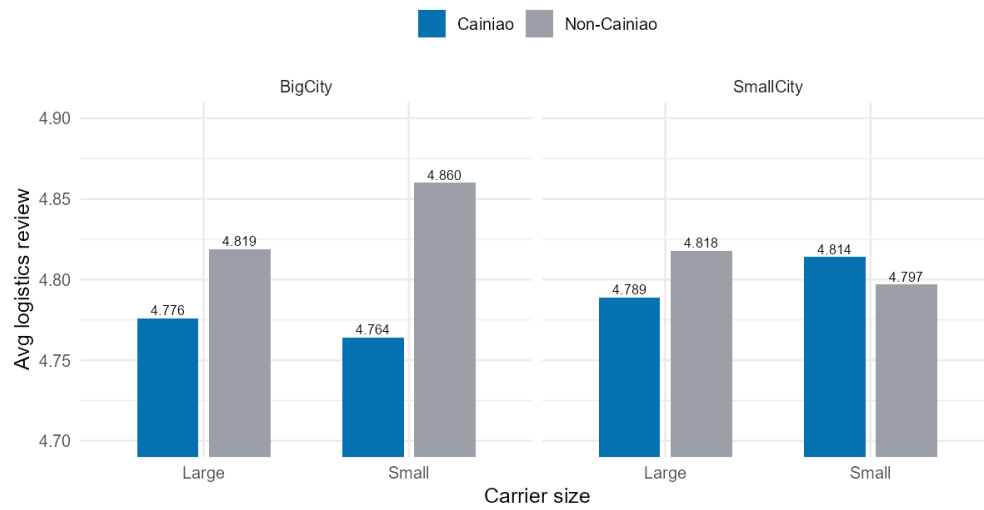


Figure 4: Customer reviews mirror the speed punchline.